

Question # 1:

1) Range = Maximum value - Minimum value
 $R = 177 - 61 = 116$

2) No. of Classes:

$$m = 1 + 3.3 \log N$$

$$= 1 + 3.3 \log(100) = 7.6$$

3) Size of Class:

$$h = R/m = 116/7.6 = 15.26$$

According to Question statement:

Index	Tally	Frequency
61-65		4
66-70		2
71-75		3
76-80		3
81-85		1
86-90		4
91-95		5
96-100		4
101-105		1
106-110		3
111-115		2
116-120		0
121-125		2

126 - 130		0
131 - 135	1	1
136 - 140	11	2
141 - 145	111	3
146 - 150		0
151 - 155	1	1
156 - 160		0
161 - 165		0
166 - 170		0
171 - 175 175	11	2
176 - 177		
176 - 180	11	2

Question # 2

Absences	No of days
0	5
1	7
2	9
3	6
4	4
5	2
6	1
7	1
28	35

1) Absent < 4
 Absences (0, 1, 2, 3)
 Total days (27)

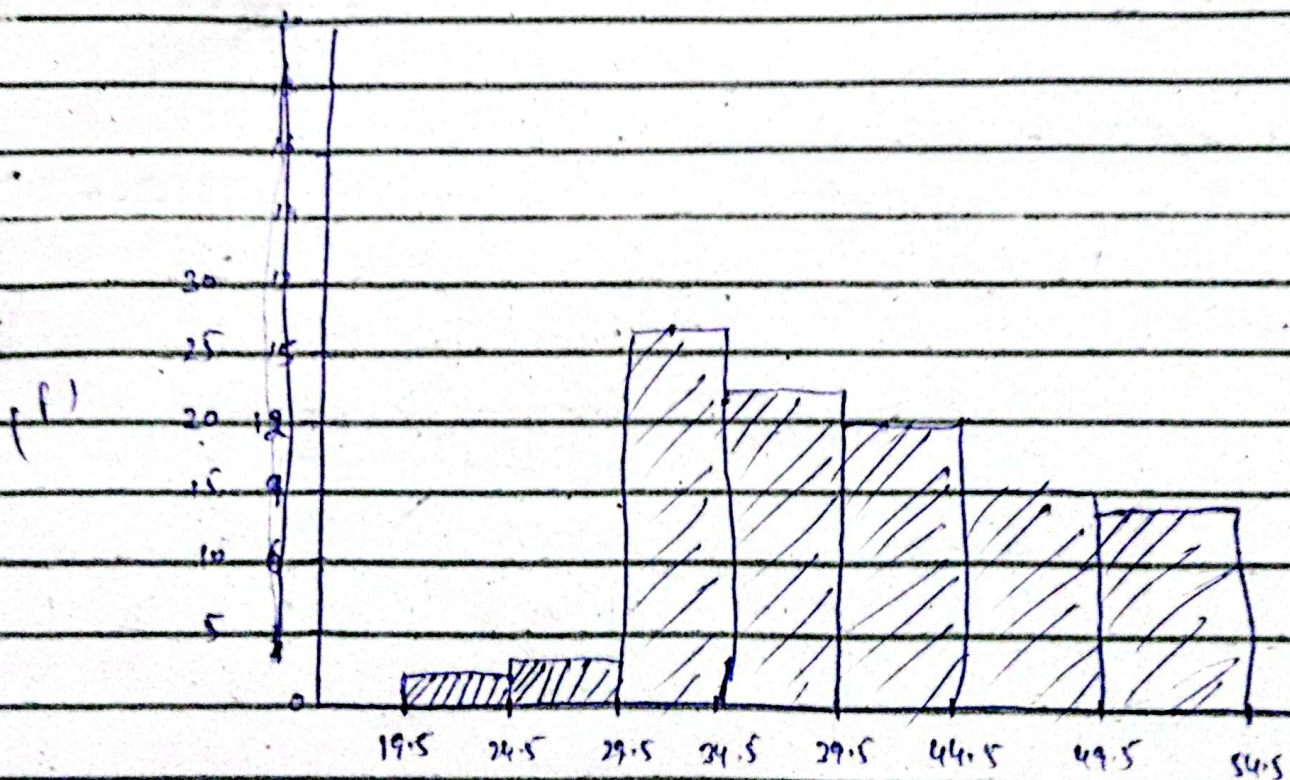
2) at least 4 people ≥ 4
 Absences (4, 5, 6, 7)
 Total day (8)

3) Total days = 35
 absences = 28

Question 3

12/2/2020

Age Nearest Birth date	Number of men (frequency)	Class Boundaries
20-24	1	19.5 - 24.5
25-29	2	24.5 - 29.5
30-34	26	29.5 - 34.5
35-39	22	34.5 - 39.5
40-44	20	39.5 - 44.5
45-49	15	44.5 - 49.5
50-54	14	49.5 - 54.5



birth date

Question # 4

frequency Distribution table.

Range = 32 - 8 = 24
 $n = 143.3 \log(32)$
 6

Class-Limits	frequency	Relative frequency	Cumulative frequency	Cum-Percent
8 - 12	5	0.25	5	25
13 - 17	1	0.05	6	30
18 - 22	3	0.15	9	45
23 - 27	6	0.30	15	75
28 - 32	5	0.25	20	100

$R = 32 - 8$
 $= 24$

20

1

$m = 5$

$\frac{5}{20} = 0.25$

$5 - 2$

$\frac{6}{20} \times 100 = 30$

$h = R/m$

$5 \times 100 = 500$

$0.30 \times 20 = 6$

$\frac{15}{20} = 75$

$24/5 = 4.8 \approx 5$

$\frac{15}{20} \times 100$

$20 - 6 = 14$

$\frac{3}{20} = 15$

$\frac{5}{20} \times 100 = 25$

0.30

20

$\frac{6}{20} \times 100 = 30$

$\frac{6}{20} \times 100 = 30$

(i) $x + 1 = 9$
 $y + 2 = 9$
 $y = 9 - 2$

$\frac{1}{20} = 0.05$

$\frac{6}{20} \times 100 = 30$

$\frac{9}{10} \times 100 = 90$

20

$\frac{6}{0.30} \times 100 = 200$

$\frac{1}{0.05} \times 100 = 200$

$\frac{6}{20} = 0.30$

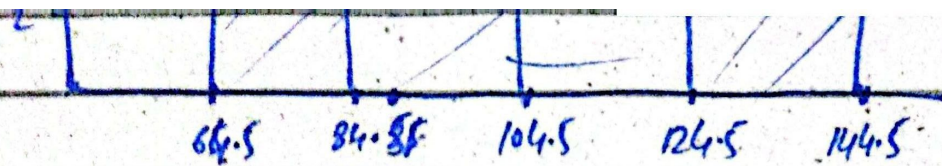
$\frac{6}{20} = 0.30$

$\frac{6}{0.30} = 20$

$\frac{1}{0.05} \times 100$

$\frac{C.F}{T.F} \times 100$

$\frac{1}{20} = 0.05$



class f_v

① frequency $\rightarrow$ Relative frequency.

$$\frac{\text{frequency}}{\text{Total frequency}} = \text{Relative frequency}$$

② Cumulative frequency $\rightarrow$ cumulative percentage

$$\frac{\text{cumulative frequency}}{\text{total frequency}} \times 100 = \text{CP}$$

③ Cumulative frequency:

$$\text{Previous frequency} + \text{Previous cumulative frequency} = \text{Cumulative frequency}$$

Question # 5

Stem	leaf	Ordered Value
0	5 7 7 9	5 7 7 9
1	0 3 5 5 7	0 3 5 5 7
2	9 0 8 4 5 2 2 4.5	0 2 2 4 4 5 5 8 9
3	0 1 0 5 5 3 5	0 0 1 3 5 5 5
4		
5		
6	3 3	3 3
7	8 9 5 0	0 5 8 9
8	0 0	0 0
9	9 0	0 9

0-2	5 7 7 9 10 13 15 15 17 20 22 22	18
	24 24 25 25 28 29	
3-5	30 30 31 33 35 35 35	7
6-9	63 63 70 75 78 79 80 80 90 99	10

Question # 6

$$n = \text{Total values} = 15$$

$$\sum x = \text{total values amount} = 13676$$

Direct Method:

$$\text{A.M.} \Rightarrow \bar{x} = \frac{\sum x}{n} = \frac{13676}{15} = 911.7$$

$$\begin{aligned} \text{G.M.} \Rightarrow (x_1 \cdot x_2 \cdot x_3 \cdots x_n)^{1/n} &= \left(\frac{13676}{15} \right)^{1/15} \\ &= (4.33 \times 10^{31})^{1/15} = 278.49 \end{aligned}$$

H.M. $\Rightarrow$

x	$1/x$	$1/x$
60	0.016	0.016
80	0.0125	0.012
90	0.0111	0.011
96		0.010
120		0.008
150		0.006
200		0.005
360		0.002
480		0.002
520		0.002
1060		0.001
1200		0.001
1450		0.001
2500		0.0004
7700		0.0001

$$\text{H.M.} = \frac{n}{\sum (1/x)}$$

$$= \frac{15}{0.0833}$$

$$= 180.07$$

Question #. 8

$$\text{Average speed} = \frac{\text{Total Distance}}{\text{Total Time}} = \frac{200}{8.7} = 23 \text{ km h}^{-1}$$

1-10

Stages	speed	distance ($\frac{200}{10}$)	time (s)
1	10	20	2
2	15	20	1.33
3	20	20	1
4	25	20	0.8
5	30	20	0.67
6	40	20	0.5
7	50	20	0.4
8	30	20	0.67
9	40	20	0.5

$$T = 280$$

$$\frac{280}{10}$$

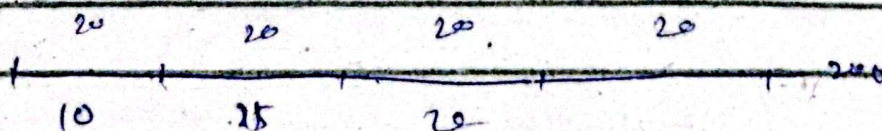
$$28$$

$$8.7$$

$$\rightarrow 31.320 \text{ s}$$

$$\text{Average Speed} = 22.9 \text{ km h}^{-1} \rightarrow 23 \text{ km h}^{-1}$$

$$\text{For every second} = 0.006 \text{ km s}^{-1}$$



Question # 9

C.F	Income	No. of workers (f)	Midpoint (x)	fx	log x	f log x	f/x	Class Boundary
15	35-39	15	37	555	1.568	23.4	0.40	34.5-39.5
28	40-44	13	42	544	1.62	21.06	0.30	39.5-44.5
45	45-49	17 f_1	47	799	1.67	28.39	0.36	44.5-49.5
✓	50-54	29 $\Rightarrow f_m$	52	1508	1.71	49.59	0.55	49.5-54.5 $h=5$
	55-59	11 f_2	57	627	1.7	18.71	0.19	54.5-59.5 $L=f_1$
	60-64	10	62	620	1.79	17.9	0.16	59.5-64.5
	65-69	6	67	402	1.82	10.72	0.08	64.5-69.5
		$\Sigma f = 101$		$\Sigma fx = 5055$		$\Sigma f \log x = 170$	$\Sigma (f/x) = 2.04$	

1) A.M:

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{5055}{101} = 50.049$$

2) G.M:

$$\left(x_1^{f_1} x_2^{f_2} \dots x_n^{f_n} \right)^{\frac{1}{\Sigma f}}$$

$$= \text{Antilog} \left[\frac{\Sigma f \log x}{\Sigma f} \right] = \text{Antilog} \left(\frac{170}{101} \right) = \text{Antilog} (1.68) = 47.86$$

3) H.M:

$$\frac{\Sigma f}{\Sigma (1/x)} = \frac{101}{2.04} = 49.50$$

4) Median:

$$L + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

$$n/2 = \frac{101}{2} = 50.5$$

$$49.5 + \frac{5}{29} (51 - 45) = 50.53$$

5) Mode:

$$L + \left[\frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \right] \times h$$

$$= 49.5 + \left[\frac{29 - 17}{(29 - 17) + (29 - 11)} \right] \times 5$$

$$= 51.5$$

6) First (Lower) and Third (Upper) Quantiles

$$Q_1 = L + \frac{h}{f} \left(\frac{n}{4} - c \right)$$

$$= 49.5 + \frac{5}{29} \left(\frac{101}{4} - 45 \right)$$

$$\frac{n}{4} = \frac{101}{4} = 25.25$$

$$Q_1 = 49.5 + \frac{5}{29} \left(\frac{101}{4} - 45 \right)$$

$$= 46.09$$

$$Q_3 = L + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

$$= 49.5 + \frac{5}{29} \left(\frac{3(101)}{4} - 45 \right)$$

$$= 54.80$$

7) Seventh Decile:

$$\begin{aligned} D_7 &= L + \frac{h}{f} \left(\frac{7n}{10} - C \right) \\ &= 49.5 + \frac{5}{29} \left(\frac{7 \times 101}{10} - 45 \right) \\ &= 53.93 \end{aligned}$$

8) Eighty-five Percentile:

$$\begin{aligned} P_{85} &= L + \frac{h}{f} \left(\frac{85n}{100} - C \right) \\ &= 56.54 \end{aligned}$$

9) Ninety-Six Percentile

$$\begin{aligned} P_{96} &= L + \frac{h}{f} \left(\frac{96n}{100} - C \right) \\ &= 58.45 \end{aligned}$$

10) Show that $Q_2 = D_5 = P_{50}$

$$\text{Median} = L + \frac{h}{f} \left(\frac{n}{2} - C \right)$$

$Q_2 = L + \frac{h}{f} \left(\frac{2n}{4} - C \right)$	$D_5 = L + \frac{h}{f} \left(\frac{5n}{10} - C \right)$	$P_{50} = L + \frac{h}{f} \left(\frac{50n}{100} - C \right)$
$= L + \frac{h}{f} \left(\frac{n}{2} - C \right)$	$= L + \frac{h}{f} \left(\frac{n}{2} - C \right)$	$= L + \frac{h}{f} \left(\frac{n}{2} - C \right)$
$= \text{Median}$	$= \text{Median}$	$= \text{Median}$

3) Mean Deviation (M.D)

Method	for ungrouped data	Grouped Data
M.D from Mean ($\bar{x}$)	$\frac{\sum x - \bar{x} }{n}$	$\frac{\sum f(x - \bar{x})}{\sum f}$
M.D from Median	$\frac{\sum x - \tilde{x} }{n}$	$\frac{\sum f(x - \tilde{x})}{\sum f}$
M.D from Mode ($\hat{x}$)	$\frac{\sum x - \hat{x} }{n}$	$\frac{\sum f(x - \hat{x})}{\sum f}$

Mean

Combined Deviation:

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + \dots}{n_1 + n_2 + \dots}$$

Combined Standard Deviation:

$$\bar{s}_c = \sqrt{\frac{n_1 (s_1^2 + d_1^2) + n_2 (s_2^2 + d_2^2) + \dots}{n_1 + n_2 + \dots}}$$

$$d_1 = \bar{x}_1 - \bar{x}_c$$

$$d_2 = \bar{x}_2 - \bar{x}_c$$

Question # 11

Show/Prove

$$n_1 = f_1 = 200$$

$$\bar{x}_1 = 25$$

$$\bar{s}_1 = 3$$

$$\bar{x}_c = 16$$

$$n_2 = f_2 = 250$$

$$\bar{x}_2 = 10$$

$$\bar{s}_2 = 4$$

$$\bar{s}_c = 7.2$$

$$n_3 = f_3 = 300$$

$$\bar{x}_3 = 15$$

$$\bar{s}_3 = 5$$

find out coefficient of variation = ?

$$\text{coefficient of variation} = \frac{\bar{s}_c}{\bar{x}_c} \times 100$$

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3}{n_1 + n_2 + n_3} = 16$$

$$\bar{s}_c = \sqrt{\frac{n_1(\bar{s}_1^2 + d_1^2) + n_2(\bar{s}_2^2 + d_2^2) + n_3(\bar{s}_3^2 + d_3^2)}{n_1 + n_2 + n_3}} = 7.2$$

$$C.V = \frac{7.2}{16} \times 100$$

$$= 45\%$$

Question # 12

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{4445}{413} = 10.76$$

Groups	frequency	x	fx	$f(x-\bar{x})$	$f(x-\bar{x})^2$	$(x-\bar{x})^3$
2-4	18	3	54	-7.76	60.21	467.28
4-6	24	5	120			
6-8	47	7	329			
8-10	80	9	720			
10-12	102	11	1122			
12-14	66	13	858			
14-16	40	15	600			
16-18	21	17	357			
18-20	15	19	285			
	413		4445			

$$m_0 = \frac{\sum f(x-\bar{x})^0}{\sum f}$$

$$m_0 = \frac{\sum f(x-\bar{x})^0}{\sum f} = 1$$

$$m_1 = \frac{\sum f(x-\bar{x})^1}{\sum f} = 0$$

$$m_2 = \frac{\sum f(x-\bar{x})^2}{\sum f} = 13.76$$

$$m_3 = \frac{\sum f(x-\bar{x})^3}{\sum f} = 3.28$$

$$m_4 = \frac{\sum f(x-\bar{x})^4}{\sum f} = 522.87$$

$$p = 2$$

$$(x-\bar{x}) = 4$$

$$b_1 = \frac{(m_3)^2}{(m_2)^3}$$

$$= \frac{(3.28)^2}{(13.76)^3} = 41295$$

$$b_2 = \frac{m_4}{(m_2)^4}$$

$$= \frac{522.87}{(13.76)^4}$$

$$= 2.76$$